

ANTON ALYAKIN

alyakin314@gmail.com ◇ alyakin314.github.io

EDUCATION

Washington University in St. Louis

Doctor of Medicine

2024 Congress of Neurological Surgeons AI Fellow

[Aug 2021, Dec 2026]

(Expected)

Johns Hopkins University

Master of Science in Engineering

Applied Mathematics & Statistics

[Jan 2019, Dec 2019]

Johns Hopkins University

Bachelor of Science

Computer Science

Applied Mathematics & Statistics

[Aug 2015, May 2019]

RESEARCH

NYU Langone Health

AI Research Engineer / Visiting Medical Student

Neurosurgery / OLAB

[Mar 2024, present]

Washington University in Saint Louis

Medical Student Researcher

Neurosurgery / Leuthardt Lab

[Mar 2023, May 2026]

Johns Hopkins University

Assistant Research Engineer

Applied Mathematics & Statistics / Neurodata Lab

[Jan 2020, Mar 2021]

Graduate Research Assistant

Applied Mathematics & Statistics / Priebe Group

[Jun 2019, Dec 2019]

Undergraduate Research Assistant

Computer Science / Saria Lab

[May 2017, May 2019]

SELECTED PUBLICATIONS

1. **A. Alyakin**, J. Stryker, D. A. Alber, J. V. Lee, K. L. Sangwon, B. Duderstadt, A. Save, D. Kurland, S. Frome, S. Singh, J. Zhang, E. Yang, K. Y. Park, C. Orillac, A. Valliani, S. Neifert, A. Liu, A. Patel, C. Livia, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, R. Feng, T. Hollon, Y. Aphinyanaphongs, J. G. Golfinos, L. Snyder, E. Leuthardt, D. Kondziolka, E. K. Oermann, *CNS-Obsidian: A Neurosurgical Vision-Language Model Built From Scientific Publications*, Neurosurgery, 2026. [journal] [arXiv] [code] [model]
2. K. Vishwanath, **A. Alyakin**, M. Ghosh, A. Hage, S. N. Neifert, C. Orillac, N. Mandelberg, H. Khan, J. V. Lee, J. Yao, W. R. Small, A. Varma, B. Hewitt, Y. Aphinyanaphongs, D. A. Alber, E. K. Oermann, *General-Purpose Large Language Models Outperform Specialized Clinical AI Tools on Medical Benchmarks*, Nature Medicine, 2026. [journal] [arXiv] [code]
3. D. A. Alber, Z. Yang, **A. Alyakin**, E. Yang, J. Zhang, S. Rai, A. A. Valliani, G. R. Rosenbaum, A. K. Amend-Thomas, D. B. Kurland, C. M. Kremer, A. Eremiev, B. Negash, D. D. Wiggan, M. A. Nakatsuka, K. L. Sangwon, S. Neifert, H. A. Khan, A. V. Save, A. Palla, E. A. Grin, M. Hedman, M. Nasir-Moin, C. X. Liu, L. Y. Jiang, M. A. Mankowski, D. L. Segev, Y. Aphinyanaphongs, H. A. Riina, J. G. Golfinos, D. A. Orringer, D. Kondziolka, E. K. Oermann, *Medical Large Language Models are Vulnerable to Data-poisoning Attacks*, Nature Medicine, 2025. [journal] [code]

OTHER PUBLICATIONS

1. **A. Alyakin**, E. K. Oermann, *Current Artificial Intelligence: Large Language Model Applications in Neurosurgery*, Neurosurgery, accepted.
2. M. Nakatsuka, D. D. Wiggan, Z. E. Morakinyo, C. Cubells, M. Murata, **A. Alyakin**, E. K. Oermann, *GPT-4-Assisted versus Human Detection of Intraoperative Adverse Events from Spine Surgery Operative Reports*, Neurosurgery Practice, accepted.
3. A. K. Kapoor, **A. Alyakin**, J. V. Lee, E. Yang, K. Vishwanath, A. M. Schulze, J. Lee, Y. Aphinyanaphongs, H. Riina, J. A. Frontera, E. K. Oermann, *Large Language Models for Predicting Functional Outcomes after Acute Ischemic Stroke*, accepted. [arXiv]
4. A. K. Kapoor, **A. Alyakin**, R. Santarosa, C. William, S. Hoda, Y. Wang, A. Garrison, I. Littig, H. Weiss, E. K. Oermann, *Hemorrhagic Synovial Cyst Mimicking a Neoplastic Process: illustrative case*, Journal of Neurosurgery: Case Lessons. [journal]
5. A. Kapoor, **A. Alyakin**, J. E. Markert, A. Arias, M. Sughrue, E. Yang, K. Vishwanath, J. V. Lee, E. K. Oermann, *Big Data in U.S. Neuro-Oncology: Trends and Translational Priorities*, Journal of Neuro-Oncology, 2026. [journal]
6. **A. Alyakin**, D. A. Alber, J. V. Lee, J. Stryker, S. Singh, A. Save, D. Kurland, C. Orillac, A. A. Valliani, S. Neifert, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, L. Snyder, E. C. Leuthardt, D. Kondziolka, E. K. Oermann, *Automated Generation and Human Evaluation of Neurosurgical Board Examination Self-Assessment Questions*, Neurosurgery Practice, 2026. [journal]
7. **A. Alyakin**, J. Stryker, J. V. Lee, R. Feng, T. Hollon, D. Kondziolka, E. K. Oermann, *AI-Powered Pipeline Transforms Neurosurgical Articles into High-Quality Graphical Abstracts*, Neurosurgery Practice, 2026. [journal]
8. K. L. Sangwon, X. Han, A. Becker, Y. Zhang, R. Ni, J. Zhang, D. A. Alber, **A. Alyakin**, M. Nakatsuka, N. Fabbri, Y. Aphinyanaphongs, J. Yang, A. Chachoua, D. Kondziolka, I. Laufer, E. K. Oermann, *Automating the Referral of Bone Metastases Patients With and Without the Use of Large Language Models*, Neurosurgery, 2026. [journal]
9. N. I. Maarouf, D. Reinecke, A. Smith, J. Markert, T. G. Cogan, X. Han, **A. Alyakin**, D. A. Alber; M. Park, N. K. Goff, H. Weiss, E. Harake, K. Eddy, T. Hollon, E. K. Oermann, D. A. Orringer, *Natural Language Processing Methods Automate Molecular Marker Extraction From Glioma Pathology Reports*, Neurosurgery, 2026. [journal]
10. S. F. Chen, **A. Alyakin**, A. Seas, E. Yang, J. Choi, J. V. Lee, A. Chen, P. Warman, R. Bitolas, R. J. Steele, D. Alber, E. K. Oermann, *LLM-assisted Systematic Review of Large Language Models in Clinical Medicine*, Nature Medicine, 2026. [journal] [code]
11. K. Vishwanath, **A. Alyakin**, M. Ghosh, J. V. Lee, D. A. Alber, K. Sangwon, D. Kondziolka, E. K. Oermann, *Evaluating the Performance and Fragility of Large Language Models on Neurosurgical Board-like Questions*, Neurosurgery, 2025. [journal] [arXiv]
12. X. Han^{†1}, **A. Alyakin**[†], S. Ciprut, C. Lapierre, J. Stryker, J. Golfinos, D. Kondziolka, E. K. Oermann, *Neuro Data Hub: A New Approach for Streamlining Medical Clinical Research*, Neurosurgery Practice, 2025. [journal]
13. K. Vishwanath, J. Stryker, **A. Alyakin**, D. A. Alber, E. K. Oermann, *MedMobile: a Mobile-sized Language Model with Clinical Capabilities*, BMJ Digital Health & AI, 2025. [journal] [arXiv] [code]
14. S. Singh, **A. Alyakin**, D. A. Alber, J. Stryker, A. P. S. Tong, K. Sangwon, N. Goff, M. de la Paz, M. Hernandez-Rovira, K. Y. Park, D. Kondziolka, E. K. Oermann, *The Pitfalls of Multiple-Choice Questions in Generative AI and Medical Education*, Scientific Reports, 2025. [journal] [arXiv]

^{1†} symbolizes equal contribution; order preserved as in manuscript.

15. **A. Alyakin**, D. Kurland, D. Alber, K. Sangwon, D. Li, A. Tsirigos, E. Leuthardt, D. Kondziolka, E. Oermann, *CNS-CLIP: Transforming a Neurosurgical Journal into a Multimodal Medical Model*, Neurosurgery, 2025. [journal]
16. R. Guennoun[†], **A. Alyakin**[†], H. Higushi, S. Demehri, *Commensal HPVs Have Evolved to Be More Immunogenic Compared with High-Risk α -HPVs*, Vaccines, 2024. [journal]
17. J. Lee, M. A. Ruiz-Cardozo, R. P. Patel, S. Javeed, R. S. Lavandi, C. Newsom-Stewart, **A. Alyakin**, C. A. Molina, N. Agarwal, W. Z. Ray, M. Santacatterina, B. H. Pennicooke, *Clinical Prediction for Surgical versus Nonsurgical Interventions in Patients with Vertebral Osteomyelitis and Discitis*, Journal of Spine Surgery, 2024. [journal]
18. **A. A. Alyakin**, J. Agterberg, H. S. Helm, and C. E. Priebe, *Correcting a Nonparametric Two-sample Graph Hypothesis Test for Graphs with Different Numbers of Vertices with Applications to Connectomics*, Applied Network Science, 2024. [journal] [arXiv] [code]
19. J. Chung[†], B. Varjavand[†], J. Arroyo, **A. Alyakin**, J. Agterberg, M. Tang, J. T. Vogelstein, C. E. Priebe, *Valid Two-Sample Graph Testing via Optimal Transport Procrustes and Multiscale Graph Correlation with Applications in Connectomics*, Stat, 2022. [journal] [arXiv] [code]
20. M. Powell, C. Clark, **A. Alyakin**, J. T. Vogelstein, B. Hart, *Exploration of Residual Confounding in Analyses of Associations of Metformin Use and Outcomes in Adults With Type 2 Diabetes*, JAMA Network Open, 2022. [journal] [arXiv]
21. F. Rahman, N. Finkelstein, **A. Alyakin**, N. A. Gilotra, J. Trost, S. P. Schulman, S. Saria, *Using Machine Learning for Early Prediction of Cardiogenic Shock in Patients with Acute Heart Failure*, Journal of the Society for Cardiovascular Angiography & Interventions, 2022. [journal] [arXiv]
22. K. Marchisio, Y. Park, A. Saad-Eldin, **A. Alyakin**, K. Duh, C. Priebe, P. Koehn, *An Analysis of Euclidean vs. Graph-Based Framing for Bilingual Lexicon Induction from Word Embedding Spaces*, Findings of the Association for Computational Linguistics: EMNLP 2021. [journal] [arXiv] [code]

PRESENTATIONS

1. S. Madira, **A. Alyakin**, M. Rahimi, J. Schilling, M. Pasque, K. Kotkar, M. Masood, R. J. Damiano Jr., T. Kaneko, A. Pawale, *Sex and Center-Level Variation in Temporary Surgically Implanted Microaxial Flow Pump Use Among Heart Transplant Candidates*, The Society of Thoracic Surgeons (STS) 63rd Annual Meeting, 2027, TBP.
2. A. McEvoy, D. Serota, R. Guennoun, **A. Alyakin**, K. Ken, F. L. Smith, A. Jambusaria-Pahlajani, L. Wheless, B. Michalski-McNeely, *Acitretin Reduces Invasive Cutaneous Squamous Cell Carcinoma Incidence in High-Risk Patients: A 16-Center Retrospective Cohort Study*, American Society for Dermatologic Surgery (ASDS) 2026 Annual Meeting, TBP.
3. J. V. Lee, E. Yang, **A. Alyakin**, A. Kapoor, X. Han, B. V. Nahed, J. Mullin, E. K. Oermann, *Contrastively Fine-Tuned Dual Encoders Automate Retrieval of CPT Codes from Operative Notes*, oral presentation, Congress of Neurological Surgeons 2026 Annual Meeting, TBP.
4. B. Michalski-McNeely, D. Serota, R. Guennoun, **A. Alyakin**, K. Ken, F. L. Smith, A. Jambusaria-Pahlajani, L. Wheless, *Acitretin for Keratinocyte Carcinoma Chemoprevention: Efficacy, Onset, and Durability*, International Immunosuppression & Transplant Skin Cancer Collaborative (ITSCC) Biennial Symposium, 2026.
5. **A. Alyakin**, J. Stryker, D. A. Alber, J. V. Lee, E. Leuthardt, D. Kondziolka, E. K. Oermann, *CNS-Obsidian: A Neurosurgery-Tailored Vision-Language Model Fine-Tuned Exclusively on Scientific Publications*, Oral Presentation at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [journal] [poster]

6. X. Han[†], **A. Alyakin**[†], S. Ciprut, C. Lapierre, J. Stryker, A. Lowe, J. Golfinos, D. Kondziolka, E. K. Oermann, *Neuro Data Hub: A New Approach For Streamlining Medical Clinical Research*, Oral Presentation at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025.
7. K. Vishwanath, **A. Alyakin**, M. Ghosh, J. V. Lee, D. A. Alber, K. Sangwon, D. Kondziolka, E. K. Oermann, *Evaluating the Performance and Fragility of Large Language Models on Neurosurgical Board-like Questions*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [journal] [poster]
8. **A. Alyakin**, *Computational Optimization Of Laser Interstitial Thermal Therapy Parameters For Brain Tumor Ablation: A Thermodynamics-Based Simulation Approach*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [journal] [poster]
9. **A. Alyakin**, J. Stryker, S. Singh, D. A. Alber, A. Save, D. Kurland, C. Orillac, A. A. Valliani, S. Neifert, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, L. Snyder, E. C. Leuthardt, D. Kondziolka, E. Oermann, *Automated Generation And Human Evaluation Of Neurosurgical Board Examination And Self-Assessment Questions*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [journal] [poster]
10. A. Y. Fu, A. A. Valliani, **A. Alyakin**, D. Kondziolka, E. K. Oermann, *Tabular Foundation Models Predict Clinical Outcomes in Skull Base and Cerebrovascular Surgery*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [journal] [poster]
11. A. Save, **A. Alyakin**, J. Stryker, D. A. Alber, J. Zhang, G. Rosenbaum, S. Singh, E. Leuthardt, D. Kondziolka, E. K. Oermann, *A Platform for Serving Large Language Models for Neurosurgical Diagnostic Support*, 93rd American Association of Neurological Surgeons Annual Scientific Meeting, 2025.
12. K. L. Sangwon, **A. Alyakin**, D. Kurland, E. Leuthardt, D. Kondziolka, E. K. Oermann, *A Generalizable Pipeline for Building an Extensive Domain-Specific Dataset from a Medical Journal - Neurosurgery Edition*, Oral Presentation at the 74th Annual Meeting of the Congress of Neurological Surgeons, 2024. [journal]
13. **A. Alyakin**, D. Kurland, D. Alber, K. Sangwon, D. Li, A. Tsirigos, E. Leuthardt, D. Kondziolka, E. Oermann, *CNS-CLIP: Transforming a Neurosurgical Journal into a Multimodal Medical Model*, Oral Presentation at the 74th Annual Meeting of the Congress of Neurological Surgeons, 2024.

POSTERS

1. A. Kapoor, J. V. Lee, **A. Alyakin**, E. Yang, J. Frontera, E. K. Oermann, *Pretrained Tabular Foundation Model Predicts Functional Outcomes After Acute Ischemic Stroke*, Digital Poster at the 76th Annual Meeting of the Congress of Neurological Surgeons, 2026.
2. J. E. Markert, N. K. Goff, **A. Alyakin**, J. V. Lee, K. Vishwanath, E. Yang, A. Kapoor, J. J. Choi, R. K. Steele, A. Schulze, S. F. Chen, J. Lee, E. K. Oermann, *Computer Vision Models In Neurosurgery: A Systematic Analysis And Readiness Assessment*, Digital Poster at the 76th Annual Meeting of the Congress of Neurological Surgeons, 2026.
3. A. K. Kapoor, **A. Alyakin**, E. Yang, A. S. Korpus, A. Schulze, J. E. Markert, J. A. Frontera, E. K. Oermann, *Predicting Functional Outcomes After Stroke Using Large Language Models*, Digital Poster at the 94th American Association of Neurological Surgeons Annual Scientific Meeting, 2026.
4. J. Choi, **A. Alyakin**, A. Schulze, S. F. Chen, E. K. Oermann, *Large Language Model-assisted Systematic Review of Generative AI in Neurosurgery*, Digital Poster at the 94th American Association of Neurological Surgeons Annual Scientific Meeting, 2026.

5. **A. Alyakin**, J. Stryker, J. V. Lee, R. Feng, T. Hollon, D. Kondziolka, E. K. Oermann, *AI-Powered Pipeline Transforms Neurosurgical Articles into High-Quality Graphical Abstracts*, Digital Poster at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [poster]
6. **A. Alyakin**, J. V. Lee, H. Riina, E. K. Oermann, *Computational Optimization Of Endovascular Coil Deployment Parameters For Aneurysm Treatment: A Physics-Based Simulation Approach*, Digital Poster at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [poster]
7. S. Singh, J. Zhang, **A. Alyakin**, E. K. Oermann, *GRPO Versus SFT: Optimizing Large Language Models For Neurosurgical Decision Support*, Physical Poster at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [poster]
8. A. Kapoor, **A. Alyakin**, L. Blinder, A. Korpus, E. Yang, R. J. Steele, A. M. Schulze, J. A. Frontera, E. K. Oermann, *Large Language Models Predict Modified Rankin Scale After Acute Ischemic Stroke*, Digital Poster at the 75th Annual Meeting of the Congress of Neurological Surgeons, 2025. [poster]
9. G. R. Rosenbaum, L. Y. Jiang, I. Sheth, J. Stryker, **A. Alyakin**, D. A. Alber, N. K. Goff, Y. J. F. Kwon, J. E. Markert, M. Nasir-Moin, J. M. Niehues, K. L. Sangwon, E. Yang, E. K. Oermann, *MedG-KRP: Medical Graph Knowledge Representation Probing*, Findings of the 4th Machine Learning for Health symposium (ML4H), 2024. [arXiv]
10. K. L. Sangwon, D. Kurland, **A. Alyakin**, D. Kondziolka, E. K. Oermann, *Seven Decades Of Change: Tracing The Evolution Of Neurosurgery Through Lexical Analysis Of Neurosurgery Publications Of The CNS (1955-2024)*, Digital Poster at the 74th Annual Meeting of the Congress of Neurological Surgeons, 2024.

PREPRINTS

1. K. Vishwanath, H. Lin, **A. Alyakin**, J. V. Lee, D. B. Hewitt, J. J. Yao, W. R. Small, H. A. Khan, C. Orillac, A. Varma, B. Ye, D. A. Alber, G. Stolovitzky, B. Wiesenfeld, O. Nov, W. Wu, K. Zhang, Y. Aphinyanaphongs, T. Requarth[†], E. K. Oermann[‡], The International Digital Twin Consortium in Healthcare and Medicine, *Clinician Use of Language Models Diverges from How the Models Are Evaluated*, submitted. [arXiv]
2. K. Vishwanath, B. Ye, **A. Alyakin**, J. E. Markert, A. Hsieh, M. Mańkowski, E. K. Oermann, *Large language models are vulnerable to incidental information in clinical documentation and reasoning*, submitted. [arXiv]
3. J. E. Markert, N. K. Goff, **A. Alyakin**, J. V. Lee, K. Vishwanath, E. Yang, A. Kapoor, J. J. Choi, R. K. Steele, A. Schulze, S. F. Chen, J. Lee, E. K. Oermann, *Computer Vision in Clinical Medicine: an LLM-powered Large-scale Systematic Review*, submitted. [preprint]
4. **A. Alyakin**, "All You Need" Is Not All You Need for a Paper Title: On the Origins of a Scientific Meme, 2025. [arXiv] [code]
5. L. Y. Jiang[†], A. Chen[†], X. Han, X. C. Liu, R. Dua, K. Eaton, F. Wolff, R. Steele, J. Zhang, **A. Alyakin**, Q. Pan, Y. Chen, K. L. Sangwon, D. A. Alber, J. Stryker, J. V. Lee, Y. Aphinyanaphongs, K. Cho, E. K. Oermann, *Generalist Foundation Models Are Not Clinical Enough for Hospital Operations*, submitted. [arXiv]
6. K. Vishwanath, **A. Alyakin**, D. A. Alber, J. V. Lee, A. Anand, D. Kondziolka, E. K. Oermann, *Medical large language models are easily distracted*, submitted. [arXiv] [code] [dataset]
7. C. Hang, R. Deng, L. Y. Jiang, Z. Yang, D. A. Alber, **A. Alyakin**, E. K. Oermann, *BPQA Dataset: Evaluating How Well Language Models Leverage Blood Pressures to Answer Biomedical Questions*, 2025. [arXiv]
8. **A. Alyakin**, Y. Qin, and C. E. Priebe, *LqRT: Robust Hypothesis Testing of Location Parameters using Lq-Likelihood-Ratio-Type Test in Python*, 2019. [arXiv] [code]

SUBMITTED

1. J. V. Lee, E. Yang, **A. Alyakin**, A. Kapoor, X. Han, B. V. Nahed, J. Mullin, E. K. Oermann, *Contrastive Fine-tuning of Asymmetric Dual Encoders for Multi-label CPT Code Retrieval from Operative Notes*, submitted.
2. A. Kapoor, J. V. Lee, **A. Alyakin**, E. Yang, J. Frontera, E. K. Oermann, *Pretrained Tabular Foundation Model Predicts Functional Outcomes After Acute Ischemic Stroke*, submitted.
3. K. Vishwanath, **A. Alyakin**, S. F. Chen, A. Hsieh, M. Mankowski, E. K. Oermann, *The Distributional Mismatch Between Medical AI Evaluations and Real Patient Populations*, submitted.
4. J. E. Markert, A. Arias, A. Kapoor, D. Kondziolka, **A. Alyakin**, E. K. Oermann, *Mapping Collaboration in Neurosurgical Artificial Intelligence: A Systematic Review and a Framework for Clinicians*, submitted.
5. A. Arias, J. Markert, **A. Alyakin**, A. Kapoor, D. Orringer, E. K. Oermann, *Predictors of Neurosurgical Academic Placement and Research Productivity: A Systematic Review*, submitted.
6. Y. Hong, S. Son, H. Jung, K. Vishwanath, **A. Alyakin**, J. V. Lee, E. K. Oermann, M. Sung, *Learning to Ignore: Enhancing Medical QA robustness under Unseen Distractors*, submitted.

DISSERTATIONS

1. **A. Alyakin**, *Robust Hypothesis Testing of Location Parameters using Lq-Likelihood-Ratio-Type Test in Python*, a thesis submitted to The Johns Hopkins University in conformity with the requirements for the degree of Master of Science in Engineering, 2019. [arXiv] [library] [code]
2. **A. Alyakin**, *Motif Discovery in the Irregularly Sampled Time Series Data*, a thesis submitted to The Johns Hopkins University in conformity with the requirements for Senior Honors Thesis in Computer Science, 2019. [osf] [pdf]

TALKS

Congress of Neurological Surgeons

CNS 2025 Special Symposium 3: Data Science and AI for Neurosurgeons Oct 11th, 2025
Vision-Language Models

2025 Artificial Intelligence in Neurosurgery: A Practical Course May 31st, 2025
How Can We Build Foundation Models with Neurosurgical Data

Global AI Frontier Lab

Global AI Frontier Lab: Seminar Series Jun 16th, 2025
Data Quality is Multidimensional: Product-Market-Data Fit in AI and Neurosurgery

TEACHING

Congress of Neurological Surgeons

Faculty

CNS 2025 Special Symposium 3: Data Science and AI for Neurosurgeons Oct 2025

2025 Artificial Intelligence in Neurosurgery: A Practical Course May 2025

Teaching Assistant

CNS 2024 Special Symposium 3: Data Science and AI for Neurosurgeons Sep 2024

Johns Hopkins University

Teaching Assistant

580.475 Biomedical Data Science Fall 2019

553.430/630 Introduction to Statistics Spring 2019

553.436/636 Data Mining Fall 2018

MENTORING

Anjali Kapoor

Post-baccalaureate Researcher at NYU OLAB

[2025, present]

3 manuscripts

Neurosurgery

Journal of Neuro-Oncology

Journal of Neurosurgery: Case Lessons

2 posters (CNS 2025, AANS 2026)

Krithik Vishwanath

Visiting Undergraduate Researcher at NYU OLAB

[2024, present]

3 manuscripts

Nature Medicine

Neurosurgery

BMJ Digital Health & AI

Churchill scholar

Shrutika Singh

Undergraduate Researcher at NYU OLAB

[2024, present]

1 manuscript

Scientific Reports

1 oral presentation (CNS 2025)

1 poster (CNS 2025)

Joanne Choi

Visiting Medical Student Researcher at NYU OLAB

[2025, 2025]

1 poster (AANS 2026)

1 *CNS-Nexus* case

Gabriel Rosenbaum

High School Student Researcher at NYU OLAB

[2024, 2024]

1 poster (ML4H 2024)

AWARDS

Johns Hopkins University

Applied Mathematics & Statistics Prize for Outstanding Master's Research

2020

Applied Mathematics & Statistics Achievement Award

2019

Undergraduate Departmental Honors with Thesis, Computer Science

2019

Undergraduate Departmental Honors, Applied Mathematics & Statistics

2019

Undergraduate General Honors

2019

Whiting School of Engineering Dean's List (8/8 Semesters)

2015-2019

TASIS The American School In Switzerland

2015 Shah Akbar Khan Award for Excellence in Mathematics

2015

2015 AP Scholar with Distinction

2015

Valedictorian

2015

SOCIETIES

Congress of Neurological Surgeons

Medical Student Member

[2024, present]

CNS Quarterly Member Spotlight

Summer 2025

Sergeant-At-Arms

CNS 2025

American Association of Neurological Surgeons

Medical Student Member

[2021, present]

GROUPS

Washington University School of Medicine

WUSM Global Health and Medicine

Treasurer [2022, 2022]

WUSM Global Surgery Student Alliance Chapter

Treasurer [2022, 2022]

WUSM ClimbRx (Climbing Club)

Co-lead [2022, 2022]

REVIEW

Neurosurgery

Reviewer [2024, present]

5 articles

Quantitative Imaging in Medicine and Surgery

Reviewer [2024, present]

2 articles

npj: Digital Medicine

Reviewer [2025, present]

3 articles

Scientific Reports

Reviewer [2025, present]

3 articles

BMC Neurology

Reviewer [2025, present]

1 article

IEEE Transactions on Signal and Information Processing over Networks

Reviewer [2026, present]

1 article

Frontiers in Artificial Intelligence

Reviewer [2026, present]

1 article

PLoS ONE

Reviewer [2026, present]

2 articles

SKILLS

Languages (in order of proficiency):

Python, English, Russian, R, Matlab.

Python skills:

PyTorch[DDP, FSDP, DeepSpeed, PyTorchLightning, LightningFabric, HuggingFaceAccelerate], Tensorflow, vLLM, W&B, API[OpenAI, Anthropic, AWS Bedrock].

Other skills:

L^AT_EX, Git, Docker, Databases[PostgreSQL, BigQuery], Agentic Engineering[ClaudeCode, Codex], HPC[IBM Spectrum LSF, SLURM], Platforms[Heroku, Fly.io, Vercel], Harnesses[Pi, OpenClaw], Lead climbing (6b/5.10), Bouldering (7a/V6 indoor; V2 outdoor).